

TASK

Fit an ARMA and/or an ARIMA model to the data (you may want to try both) and use it to predict the values for the six 3-month periods from January 2022 to April 2023.

SOLUTION

To address the task, three models (one ARMA model and two ARIMA models) were used to fit the data. The model with the best accuracy score was chosen to make predictions.

Autocorrelation function and Partial Autocorrelation function were plotted to see the interaction between the Autoregressive (AR) component and the moving average (MA) component in the data. The ACF and PACF plots showed slow decay. The order of both the moving average component and the autoregressive component cannot be determined for the ARMA model using the ACF and PACF plots.

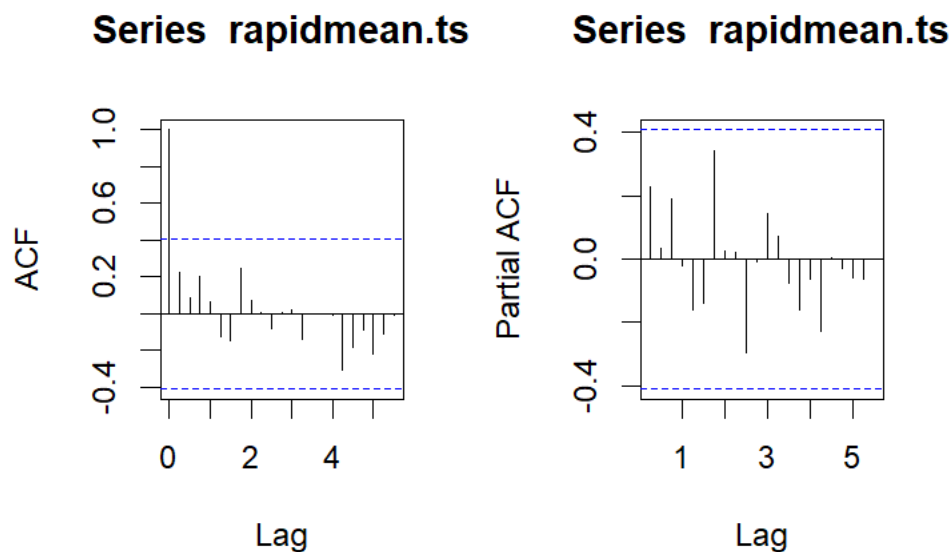


Fig. 1. The ACF and PACF plots

AKAIKE INFORMATION CRITERION(AIC)/BAYESIAN INFORMATION CRITERION(BIC)

To determine the order of the ARMA model, the AIC and BIC values of series of models with different MA and AR orders were determined and the model with the lowest AIC or BIC value was selected. For this task AIC is prioritized over BIC as AIC focuses on model that minimizes prediction error.

ARMA MODEL

Arma model with AR (1) and MA (3) was selected as the best model with the lowest AIC value. After fitting the data with the ARMA model, residual diagnostics were carried out to see if the model fits the data appropriately. The Ljung-Box test for the residuals has a p-value of 0.3164 meaning that the residuals are not correlated, and their distribution resembles a normal distribution. The predictive accuracy of the ARMA model was also measured with different accuracy scores.

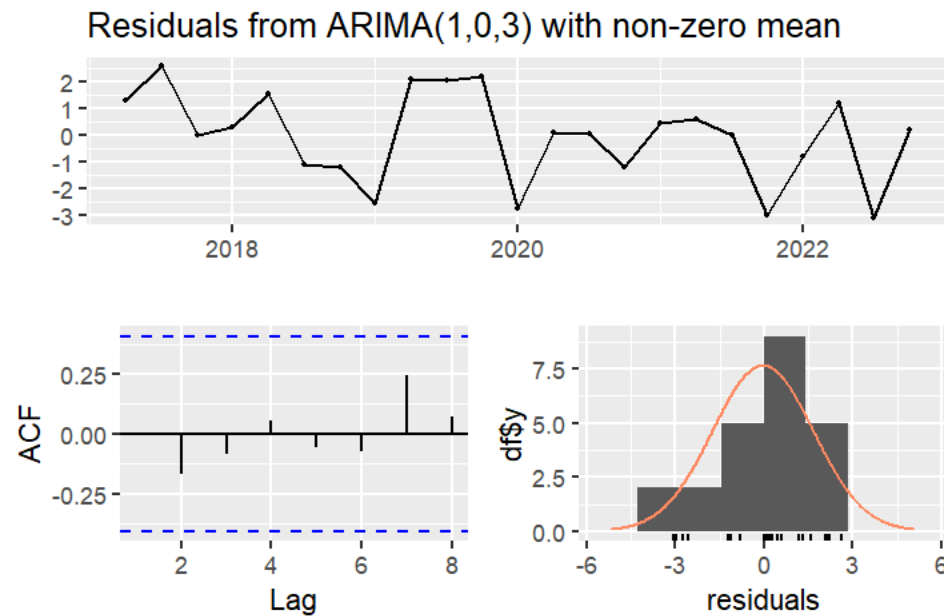


Fig 2. Showing the plots of the residuals, the residuals autocorrelation function plot and histogram plot of the residuals

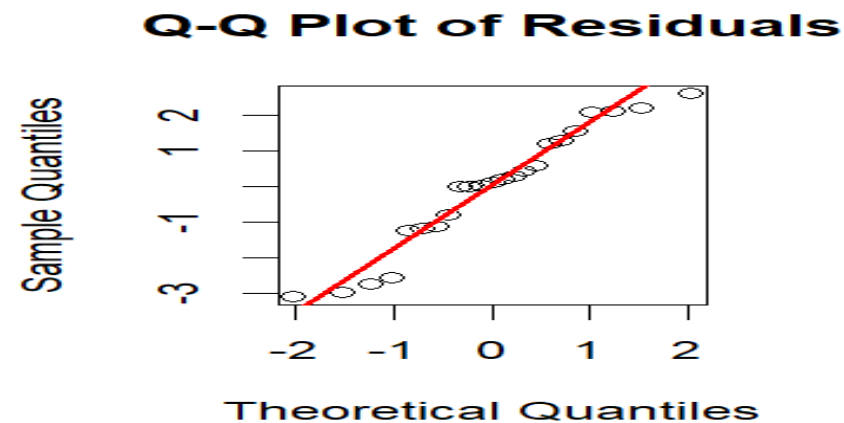


Fig 3. Showing the QQ plots of the residuals

Table 1. Accuracy Metrics for the ARMA model

ME	RMSE	MAE	MPE	MAPE
-0.04315605	1.668719	1.323077	-1.376749	8.41146

Where ME is the Mean Error, RMSE is the root mean square error, MAE is the mean absolute error, MPE is the mean percentage error, MAPE is the mean absolute percentage error

ARIMA MODEL

An augmented Dickey Fuller test was carried out to check if the data is non-stationary. The p-value of the test 0.3846 means that the null hypothesis, that is, the data is not stationary cannot be rejected. Since this is assumed to be a non-stationary from the test, differencing was done to make the data stationary. After the differencing was performed, the augmented Dickey Fuller test was carried out again to confirm that the differenced data is now stationary. The p-value(0.045) of the test showed that the differenced data is now stationary.

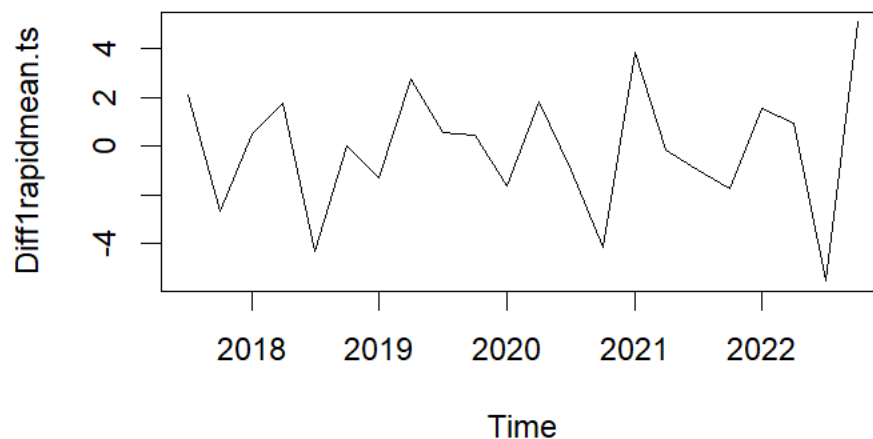


Fig 4. Showing the differenced data

To determine the order of the AR component and MA component of the ARIMA model, the AIC and BIC values of models with different MA and AR orders fitted on the differenced data were determined and the model with the lowest AIC or BIC value was selected as before. ARIMA model with AR (1), d(1) and MA(3) was selected as the best model with the lowest AIC value. After fitting the rapidmean.ts data with the ARIMA model, residual diagnostics was carried out to see if the model fits the data appropriately. The Ljung-Box test for the residuals has a p-value of 0.6436 meaning that the residuals are not correlated, and their distribution also resembles a normal

distribution. The predictive accuracy of the ARIMA model was also measured with different accuracy scores.

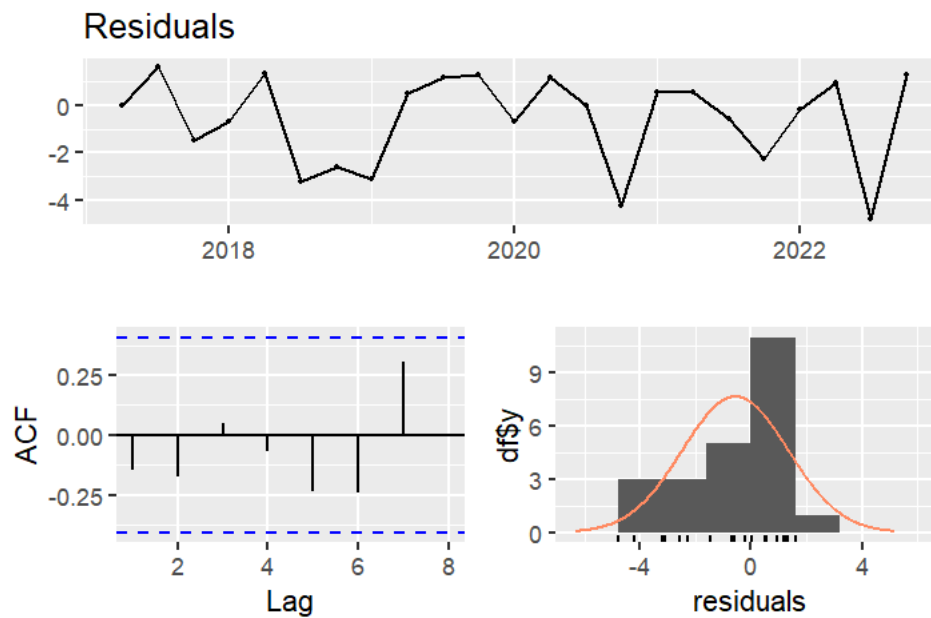


Fig 5. Showing the plots of the residuals, the residuals autocorrelation function plot and histogram plot of the residuals for ARIMA (1,1,1) model.

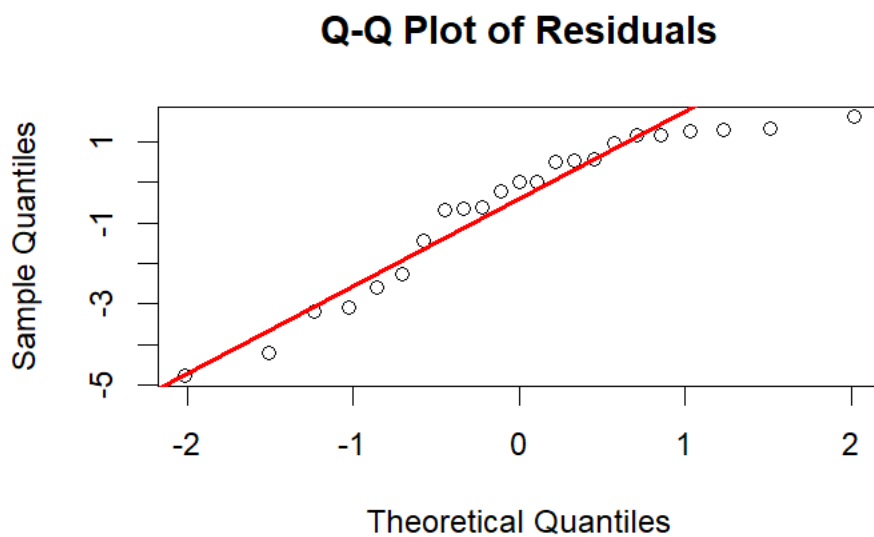


Fig 6. Showing the QQ plots of the residuals of the ARIMA (1,1,1) model

Table 2. Accuracy Metrics for the ARIMA (1,1,1) model

ME	RMSE	MAE	MPE	MAPE
-0.5729484	1.952949	1.486823	-4.864338	9.938975

THE R's AUTO-ARIMA MODEL

To further explore alternative ARIMA models, the R auto-arima function was also used to find another ARIMA models that can also fit the data. The function produced ARIMA (2,1,0) model-that is AR component of order 2 and first-order differencing. Auto-arima function automatically chooses the model by evaluating different models using the Bayesian Information Criterion(AIC) and Akaike Information Criterion. The model was then fitted on the data and the residual diagnostics was carried out.

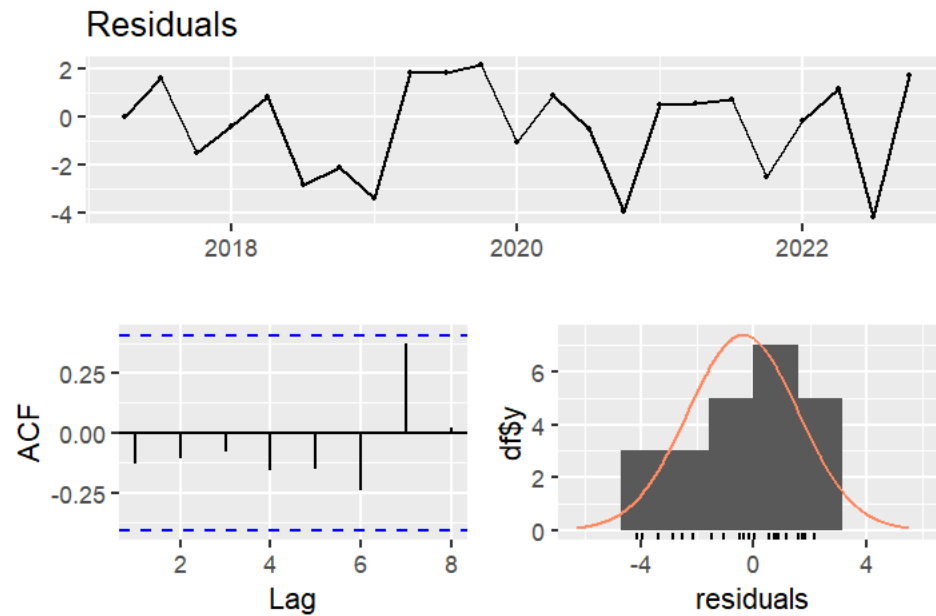


Fig 7. Showing the plots of the residuals, the residuals autocorrelation function plot and histogram plot of the residuals for ARIMA (2,1,0) model.

The Ljung-Box test for the residuals has a p-value of 0.7942 meaning that the residuals are not correlated, and their distribution also resembles a normal distribution. The predictive accuracy of the ARIMA model was also measured with different accuracy scores.

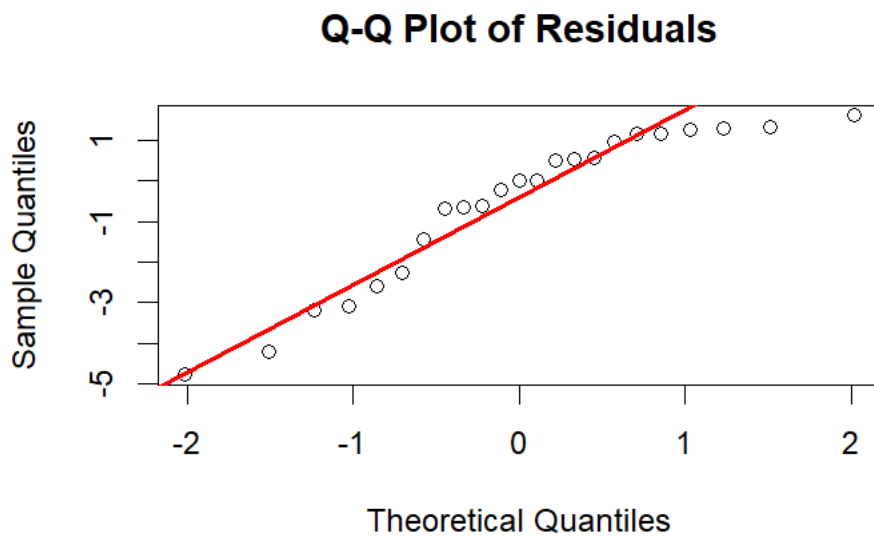


Fig 8. Showing the QQ plots of the residuals of the ARIMA (2,1,0) model

Table 3. Accuracy Metrics for the ARIMA (2,1,0) model

ME	RMSE	MAE	MPE	MAPE
-0.3751812	1.95057	1.579677	-3.573669	10.4055

MAKING PREDICTIONS FOR THE SIX 3-MONTH PERIODS

To make the predictions for the six 3-month periods between January 2022 and April 2023, the model out of the three fitted models with the best performance across the different accuracy metrics was used. Generally, based on the residual diagnostics and the accuracy scores, all the three models showed good fit to the data and their residuals are independent and resemble white noise. But the ARMA (1,0,3) model has the best accuracy scores across the performance metrics and was used for the prediction. Since the period January 2022 to February 2023 representing the first five quarters (quarter 4 of 2021 to quarter 4 of 2022 as represented in the time series data rapidmean.ts) already exist in the data used for fitting the model, their predicted values can be extracted from the fitted values of the fitted model. Then the sixth quarter, which starts from April and will be quarter 1 in 2023 will then be forecast as an unseen data. See table below for the predictions for the six 3-month data

Table 4. Comparison of the models' performances

	ME	RMSE	MAE	MPE	MAPE
ARMA	-0.04315605	1.668719	1.323077	-1.376749	8.41146
ARIMA(1,1,1)	-0.5729484	1.952949	1.486823	-4.864338	9.938975
ARIMA(2,1,0)	-0.3751812	1.95057	1.579677	-3.573669	10.4055

Table 5. Predictions Using the ARMA(1,0,3) model

	Quarter1(April-July)	Quarter 2 (August-November)	Quarter 3(December)	Quarter (January-March)
2021				17.12411
2022	16.45454	15.41780	14.18366	16.18385
2023	16.523			

APPENDIX-CODE

```
library("tseries")
library(forecast)

#Load the data to R
load("C:/Users/USER/Downloads/mocdata.RData")

View(rapid)

# Now take quarterly means (the dlm code does not work for the whole dataset - too many points)

rapid$Quarter<-floor(rapid$MM/4)+2
rapid$qyyyyy <- paste(rapid$YY,rapid$Quarter, sep='-')
rapidmean <- tapply(rapid$moc,rapid$qyyyyy,mean)

# Now convert it to a time series object. This is the data you should work with
rapidmean.ts <- ts(as.vector(rapidmean),start=c(2017,2),frequency = 4)
```

```
plot(rapidmean.ts)
```

```
rapidmean.ts
```

```
#Plot ACF and PACF for the timeseries data.
```

```
par(mfrow=c(1,2))
```

```
acf(rapidmean.ts,lag.max=30)
```

```
pacf(rapidmean.ts,lag.max=30)
```

```
#use the AIC/BIC criterion to determine the order of the ARMA process
```

```
P_order_values <- 1:3 # AutoRegressive order
```

```
Q_order_values <- 1:3 # Moving Average order
```

```
Akaike_Criterion_values <- matrix(NA, nrow = length(P_order_values), ncol =  
length(Q_order_values))
```

```
Bayesian_Criterion_Values <- matrix(NA, nrow = length(P_order_values), ncol =  
length(Q_order_values))
```

```
for (p in P_order_values) {
```

```
  for (q in Q_order_values) {
```

```
    fit <- arima(rapidmean.ts, order = c(p, 0, q))
```

```
    Akaike_Criterion_values [p , q ] <- AIC(fit)
```

```
    Bayesian_Criterion_Values [p , q ] <- BIC(fit)
```

```
  }
```

```
}
```

```
print("AIC values for models with different values of p and q :")
```

```
print(Akaike_Criterion_values)
```

```

print("AIC values for models with different values of p and q :")

print(Bayesian_Criterion_Values)


best_aic <- which(Akaike_Criterion_values == min(Akaike_Criterion_values), arr.ind = TRUE)

best_bic <- which(Bayesian_Criterion_Values == min(Bayesian_Criterion_Values), arr.ind = TRUE)


cat("Best ARMA model using Akaike Information Criteria: AR(", best_aic[1] , "), MA(", best_aic[2],
")\n", sep="")

cat("Best ARMA model using Bayesian Information Criteria:: AR(", best_bic[1], "), MA(", best_bic[2],
")\n", sep="")


#fit arma model on the data using the best found order for AR and MA

fit_arma <- arima(rapidmean.ts, order = c(1, 0, 3))


#carry out residuals diagnostics for the ARMA model

checkresiduals(fit_arma)

# Q-Q plot for the residual of the ARMA model

residuals<-residuals(fit_arma)

qqnorm(residuals, main = "Q-Q Plot of ARMA Residuals")

qqline(residuals, col = "red", lwd = 2)


#store the fitted values of the ARMA model, calculate accuracy Metrics

#for the model

fitted_Arima <- fitted(fit_arma)

accuracy_fitted_Arima <- accuracy(fitted_Arima, rapidmean.ts)

View(accuracy_fitted_Arima)

```

```
#check for stationarity
```

```
adf.test(rapidmean.ts, alternative = "stationary")
```

```
#carry out Differencing since the data is non-stationary
```

```
Diff1rapidmean.ts<-diff(rapidmean.ts, differences=1)
```

```
plot(Diff1rapidmean.ts)
```

```
#check if differenced data is now stationary
```

```
adf.test(Diff1rapidmean.ts, alternative = "stationary")
```

```
#Now estimate the best order using the differenced data
```

```
P_order_values <- 1:3 # AutoRegressive order
```

```
Q_order_values <- 1:3 # Moving Average order
```

```
Akaike_Criterion_values <- matrix(NA, nrow = length(P_order_values), ncol =  
length(Q_order_values))
```

```
Bayesian_Criterion_Values <- matrix(NA, nrow = length(P_order_values), ncol =  
length(Q_order_values))
```

```
for (p in P_order_values) {
```

```
  for (q in Q_order_values) {
```

```
    fit <- arima(Diff1rapidmean.ts, order = c(p, 0, q))
```

```
    Akaike_Criterion_values [p , q ] <- AIC(fit)
```

```
    Bayesian_Criterion_Values [p , q ] <- BIC(fit)
```

```
  }
```

```
}
```

```
print("AIC values for models with different values of p and q :")
```

```
print(Akaike_Criterion_values)
```

```
print("AIC values for models with different values of p and q :")
```

```
print(Bayesian_Criterion_Values)
```

```
best_aic <- which(Akaike_Criterion_values == min(Akaike_Criterion_values), arr.ind = TRUE)
```

```
best_bic <- which(Bayesian_Criterion_Values == min(Bayesian_Criterion_Values), arr.ind = TRUE)
```

```
cat("Best ARMA model using Akaike Information Criteria: AR(", best_aic[1], "), MA(", best_aic[2],  
")\n", sep="")
```

```
cat("Best ARMA model using Bayesian Information Criteria:: AR(", best_bic[1], "), MA(", best_bic[2],  
")\n", sep="")
```

```
#FIT ARIMA MODEL ON THE NON-STATIONARY DATA
```

```
fit_arima2 <- arima(rapidmean.ts, order = c(1, 1, 1))
```

```
residuals2<-residuals(fit_arima2)
```

```
#Perform Residual Diagnostics for the ARIMA model
```

```
checkresiduals(residuals2)
```

```
# Q-Q plot for residuals of the ARIMA MODEL
```

```
qqnorm(residuals2, main = "Q-Q Plot of ARIMA (1,1,1) Residuals")
```

```
qqline(residuals2, col = "red", lwd = 2)
```

```
#store the fitted values of the ARIMA model, calculate accuracy Metrics
```

```
#for the model
```

```
fitted_Arima2<-fitted(fit_arima2)
```

```
accuracy_fitted_Arima2 <- accuracy(fitted_Arima2, rapidmean.ts)
```

```
View(accuracy_fitted_Arima2)
```

```
#Experiment with R's auto.arima function to fit a second ARIMA model
```

```
fit_arima3 <- auto.arima(rapidmean.ts)
```

```
residuals3<-residuals(fit_arima3)
```

```
#Perform Residual Diagnostics for the second ARIMA model
```

```
checkresiduals(residuals3)
```

```
# Q-Q plot for residuals of the second ARIMA model
```

```
qqnorm(residuals3, main = "Q-Q Plot of ARIMA (2,0,1) Residuals")
```

```
qqline(residuals3, col = "red", lwd = 2)
```

```
#store the fitted values of the second ARIMA model, calculate accuracy Metrics
```

```
#for the model
```

```
fitted_Arima3<-fitted(fit_arima3)
```

```
accuracy_fitted_Arima3 <- accuracy(fitted_Arima3, rapidmean.ts)
```

```
View(accuracy_fitted_Arima3)
```

```
#produce predictions for the six quarters from January-2022 to April 2023
```

```
#using the ARMA model
```

```
#Get predictions for the first five quarters (January 2022 to March 2023) #from the fitted values.
```

```
#From the given rapidmean.ts code, first quarter starts in April
```

```
#and 4th quarter starts in January
```

```
historical_start <- c(2021, 4)
```

```
historical_end <- c(2022, 4)

historical_predictions <- window(fitted_Arima, start = historical_start, end = historical_end)

#forecast the value of the unseen quarter(quarter 1, starting in April)

future_forecast <- forecast(fit_arima, h = forecast_horizon)

FIT_ARIMA_predictions <- c(historical_predictions, future_forecast$mean)


#create a time series object with the predictions

predictions_ts <- ts(

FIT_ARIMA_predictions, start = start(historical_range),

frequency = frequency(rapidmean.ts)

)
```